Md. Shihab Uddin

Data Science | Machine Learning | AI

Dhaka, Bangladesh +880 1568945816 mdshihabuddin579@gmail.com
linkedin.com/in/mdshihab26 github.com/mdshihab-cyber portfolio.com

Professional Summary

Data Science and AI-focused undergraduate with a strong foundation in statistics, probability, linear algebra, calculus, and machine learning. Interested in developing reliable, explainable, and production-oriented AI systems. Experienced in academic research, technical projects, data analysis, and collaborative problem solving. Currently seeking opportunities in Data Science, Machine Learning, and AI engineering.

Education

B.Sc. in Computer Science and Engineering Expected 2027
American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*
CGPA: 3.84/4.00

Higher Secondary Certificate (HSC) 2022
Uttara High School and College *Dhaka, Bangladesh*
GPA: 5.00/5.00

Secondary School Certificate (SSC) 2020
Kurmitola High School and College *Dhaka, Bangladesh*
GPA: 5.00/5.00

Honors & Awards

- **Dean's List Honors** - Recognized for academic excellence and outstanding academic performance, American International University-Bangladesh (AIUB).
- **The Duke of Edinburgh's International Award – Gold Award** *In Progress — Expected March 2027*
  Undertaking the 78-week Gold level programme, involving Volunteering, Physical Recreation, Skills, an Adventurous Journey, and a Residential Project. Developing leadership, resilience, teamwork, and long-term commitment through sustained community service, skill development, and team-based challenges.
- **60% Academic Scholarship** - Awarded for outstanding academic performance and maintaining a strong CGPA for three consecutive years, American International University-Bangladesh (AIUB).

Technical Skills

Programming	Python, C/C++, JAVA, R, SQL
Data Science	Pandas, NumPy, Matplotlib
Machine Learning	Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Optimization
Deep Learning	Neural Networks, PyTorch
Data Visualization	Matplotlib, Seaborn, Plotly, Power BI
Databases	MySQL, PostgreSQL
Tools	Git, GitHub, Jupyter Notebook, Google Colab, VS Code
Research	Literature Review, Experimental Design, Statistical Analysis, Scientific Writing

Projects

Electronic Parts Manufacturer Inventory Management System Summer 2024
Java, OOP, Java Swing, File I/O

- Developed a GUI-based desktop application to streamline inventory tracking, lot number management, and expiration date monitoring for electronic parts manufacturers.
- Architected the main system structure and implemented core CRUD operations, user authentication, and data persistence using Java Swing and File I/O.
- Collaborated in a 4-person team, contributing 30% of the codebase with a primary focus on GUI development and system logic.
- GitHub Repository

Dark-Light Sensor with OLED Display Fall 2024
Arduino (C++), LDR Sensor, I2C OLED, Circuit Design

- Built an embedded system to measure real-time ambient light intensity using an Arduino UNO and a Light Dependent Resistor (LDR).

- Processed analog sensor data to compute light percentage, implementing a threshold-based algorithm to classify conditions as "Dark" or "Light" and trigger an LED indicator.
- Integrated an I2C OLED display (SSD1306) for real-time data visualization and validated circuit logic via CirKit Designer simulation before hardware prototyping.

IT Services Request Management System Fall 2024
Oracle SQL, Java (JDBC), ER Modeling, Database Design

- Designed the Entity-Relationship (ER) diagram and relational schema, applying normalization techniques (up to 3NF) to eliminate data redundancy and ensure data integrity.
- Executed complex SQL queries (Joins, Subqueries, Views, Group Functions) to extract, aggregate, and analyze service request trends, technician workloads, and departmental data.
- Integrated the Oracle database with a Java application via JDBC and implemented role-based access control (RBAC) for secure data retrieval across 3 user roles.

Real-Time Logic Gate Implementation Using Arduino Nano Spring 2025
Arduino (C++), Embedded Systems, Digital Logic, I2C

- Developed a real-time hardware system using an Arduino Nano to implement and visualize all 7 basic logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR).
- Processed analog potentiometer inputs (0–1023) to dynamically select gate types, while managing digital I/O for truth table validation and displaying real-time states on an I2C OLED screen.
- Validated system accuracy by comparing hardware outputs against standard truth tables, demonstrating the practical implementation of Boolean logic.

Stayfinder – Accommodation Booking Platform 2024 – 2025
C#, .NET, SQL, OOP, Relational Database Design

- Designed and developed a desktop-based booking platform connecting customers, hosts, and administrators, integrating a relational database with a C# GUI environment.
- Architected the database schema (8+ tables) and ER diagram, implementing complex relationships (one-to-many, many-to-many) to manage properties, bookings, payments, reviews, and user roles.
- Implemented CRUD operations, advanced search filters (location, price, amenities), and automated transaction logging to calculate platform commission (10%).
- GitHub Repository

GoldSecure – A Digital Gold Marketplace System Fall 2025
Agile/Scrum, Software Testing, Data Validation, Figma

- Collaborated in a 4-person Agile team to design and develop a secure digital marketplace, focusing on robust user data management and role-based access control.
- Designed and executed 10+ test cases (OTP verification, KYC document uploads, user registration), ensuring rigorous data validation and system reliability.
- Validated complex data flows and transaction histories to ensure accuracy, security compliance, and integrity of sensitive financial data.
- GitHub Repository

Sorting and Searching Algorithms Visualizer Spring 2025
C++, OpenGL, GLUT, Data Structures, OOP

- Collaborated in a 4-person team to build an interactive OpenGL application that visually demonstrates the step-by-step execution of 6 sorting and 2 searching algorithms.
- Implemented core algorithms (Quick, Merge, Heap, Bubble, Selection, Insertion, Binary, Linear) using Object-Oriented Design principles like inheritance and polymorphism for modular scene management.
- Utilized real-time animation loops and 2D graphics primitives (DDA, Bresenham, Cohen-Sutherland clipping) to provide visual feedback on algorithm complexity and data flow.
- GitHub Repository

Online Car Rental System Spring 2026
PHP, MySQL, AJAX/JSON, MVC, Web Security

- Developed a multi-tier web application using PHP and MySQL to manage car rentals, implementing role-based access control and end-to-end order processing.

- Designed the relational database schema (5+ tables) and executed complex SQL queries with JOINs to generate dynamic rental history, status filtering, and business analytics.
- Built AJAX/JSON API endpoints for real-time cost calculations and order placement, and ensured data integrity using PDO transactions for atomic database operations.
- Implemented robust data security measures including prepared statements (SQL injection defense), CSRF tokens, and IDOR protections to safeguard sensitive user and transaction data.

Research Experience

Researcher 2026 – Present
Ubiquitous Computer Vision and Machine Learning Research Lab (UCVML Lab) *AIUB, Dhaka, Bangladesh*

- Conducting research in artificial intelligence, machine learning, and intelligent systems under the UCVML Lab.
- Contributing to research projects involving machine learning, explainable AI, data science, and emerging AI technologies.
- Designing experiments, analyzing datasets, evaluating machine learning models, and preparing research findings for academic publication.
- Collaborating with researchers to identify research gaps, develop methodologies, and communicate experimental results through scientific writing and presentations.

Research Publications

- **Automatic Threshold Discovery in Lipid Ratios for Diabetes Risk Using Explainable ML.** *3rd International Conference on Advanced Computing Technologies (ICACT 2026)*, Kathmandu, Nepal. (IEEE Scopus Indexed)
- **StratumDP: A Method for Generating Privacy-Preserving Synthetic Data That Doesn't Erase Small Groups.** *4th International Conference on Computing Advancements (ICCA 2026)*, Dhaka, Bangladesh.
- **A Hybrid Statistical–Machine Learning Framework for Anomaly Detection in Emerging Financial Markets Using Multi-Scale Entropy and Gradient Boosting.** *2nd International Conference on Business Innovation for Inclusive Development*, Chattogram, Bangladesh
- **NeoTrust-Q: An Autonomous AI-Quantum-Resilient Cyber Defense Grid with Adaptive Zero Trust and Self-Healing Intelligence.** *3rd International Conference on Advanced Computing Technologies (ICACT 2026)*, Kathmandu, Nepal. (IEEE Scopus Indexed)
- **Why Fixed-Weight Fusion Fails: A Non-Circular Evaluation of LSTM–DBSCAN–XGBoost for Predictive Maintenance.** *11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering*, Cox's Bazar, Bangladesh. (IEEE Scopus Indexed)

Leadership & Extracurricular Activities

IEEE AIUB Student Branch Jan 2025 – Present
Dhaka, Bangladesh

- **Membership Development Coordinator** (Feb 2026 – Present): Contributing to membership growth, student engagement, and professional development initiatives.
- **Lead Volunteer** (Jan 2025 – Jan 2026): Coordinated volunteers and supported the planning and execution of technical events, workshops, and student activities.

Bangladesh Space Research Society Jun 2025 – Present
Dhaka, Bangladesh

- **Office Secretary**: Supported organizational, administrative, and coordination activities and contributed to society initiatives related to space science and technology.

Additional Information

Languages	English, Bangla, Hindi
Interests	Artificial Intelligence, Machine Learning, Research, Technology, Travel
Availability	Available for internships / research opportunities